CRAFT: Single-Example Level Generation via Context-Aware Representation and Adaptive Frequency Training

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***Abstract*—Single-example level generation addresses the critical data scarcity issue in Procedural Content Generation (PCG) by learning internal distribution from a single reference. While current methods demonstrate proficiency in capturing global structure, they often struggle with the long-tailed distribution of semantic tokens. This limitation results in the loss of details composed of low-frequency tokens, which are crucial for game play mechanics. Furthermore, current encoder relies on statistical co-occurrence while neglecting spatial context, introducing semantic ambiguity that severely degrades downstream generative performance. To address these challenges, we propose CRAFT, a diffusion-based framework tailored for 3D token-based level generation. Our method incorporates two key innovations: (1) A Voxel Variational Autoencoder (VAE) that employs multi-scale dilated convolutions to represent levels with a context-aware latent embedding, effectively resolving semantic ambiguities and enhancing downstream task performance; (2) A Dual-Phase Curriculum Learning strategy that partitions the training process into two phases—utilizing MSE loss for global structure learning and subsequently integrating an adaptively frequency re-weighted loss for detail refinement—thereby the model learns to reconstruct low-frequency tokens. Extensive experiments demonstrate that CRAFT significantly outperforms baselines, achieving a superior balance between global structural consistency and detail reconstruction. The source code is available at https://anonymous.4open.science/r/CRAFT-629C/**

***Index Terms*—Procedural content generation, Diffusion models, Variational autoencoder, Curriculum learning, Minecraft**

I. Introduction

Three-dimensional content creation tools are fundamental to modern game development. However, conventional manual pipelines are often labor-intensive, leading to high costs where content production and level design frequently dominate the total development budget [1], [2]. Even with the integration of generative AI, the persistent need for manual refinement of raw AI outputs has left this efficiency bottleneck largely unresolved [3]. As a critical downstream task of game studies, procedural level generation leverages algorithms to automate the creation of reasonable game levels. Advancing this field is essential for the rapid, large-scale generation of 3D game levels.

In video games, a level is the core component of the gaming experience, defined by the spatial layout of game entities. These layouts are organized according to topological structures that ensure playability and design intent. Our research centers on 3D level generation, adopting a token-based representation scheme to model the game levels. In this case, a level is discretized into a voxel array, where each voxel is assigned a specific class label denoting the game entity situated within that voxel. Drawing an analogy to Natural Language Processing (NLP), the set of all unique class labels is defined as a vocabulary. Each labeled voxel is mapped to a token via its corresponding integer index in the vocabulary, thereby allowing the entire 3D level to be represented as a 3D integer array.

In token-based level generation, traditional algorithms, which are based on search, grammar, or rules, often require extensive domain-specific engineering. To address these limitations, approaches based on deep learning has emerged. These approaches can automatically learn complex patterns. However, two fundamental obstacles persist for deep learning-based methods: limited data availability and content generation incoherence [4], [5]. Given that most high-quality game levels are manually crafted, acquiring large datasets containing high-quality levels remains difficult, which hinders the performance of data-sensitive models. To address this, one of approaches have pivoted toward single-example generation, aiming to train generative models on a single level. Early successes in this domain were driven by Generative Adversarial Networks (GANs), with models like TOAD-GAN [6] and World-GAN [7] demonstrating impressive capabilities in generating levels by adopting pyramid-shaped GANs with multi-scale training.However, errors accumulate during generation in multi scales [8], leading to numerous low-quality samples. Todd et al. [9] leveraged Large Language Models (LLMs) to generate levels for the game Sokoban; however, their approach was primarily restricted to small-scale 2D grids. More recently, Diffusion Models have emerged in the field.Dai et al. [4] presented DiffusionCraft, a diffusion-based framework which generates levels with fewer artifacts than GAN-based approaches.However, existing models struggle to reconstruct low-frequency tokens due to long-tailed distribution of tokens. Usually, low-frequency tokens are crucial to the gameplay loop; for example, loot containers. Therefore, this limitation affects the playability. Furthermore, existing evaluation metrics, which are calculated based on the frequency of local patches in the level, often fail to reflect true level quality because the metrics are dominated by patterns composed of high-frequency tokens, obscuring the impact of patterns composed of low-frequency tokens.

In this paper, we propose a diffusion-based method to overcome the limitations associated with the long-tailed distribution of tokens, alongside a suite of metrics to comprehensively assess the quality of generated levels. In addition, enabled by the extensive context window, few-shot learning ability and spatial reasoning ability of modern LLMs, we explore the ability of LLMs to generate large-scale 3D token-based levels. We identify four major challenges in this task. The first challenge lies in determining the optimal representation space for the diffusion process. Recent works map the one-hot encoded training level onto a high-dimensional embedding space using semantic token labels, utilizing the Block2Vec [7] method adapted from NLP. However, this method relies on statistical co-occurrence and often fails to learn meaningful representations for low-frequency tokens. To address this, we employ our proposed Voxel VAE, a Variational Autoencoder (VAE) with a multi-scale dilated convolutional encoder designed to capture the spatial contexts of voxels. The second challenge arises from the standard diffusion loss function—Mean Squared Error (MSE)—where gradients are dominated by high-frequency tokens. This imbalance causes the model to ignore low-frequency tokens, leading to poor reconstruction of rare but essential game entities. To address this, we design a Dual-Phase Curriculum Learning strategy. We adopt a two-phase training approach: the first phase employs a basic MSE loss, while the second phase progressively integrates Inverse Class Frequency (ICF) weighting into the loss function to balance the gradient contribution between high-frequency and low-frequency tokens. This approach enables diffusion models to effectively reconstruct low-frequency tokens while preserving global structural consistency. The third challenge pertains to evaluation. Existing metrics often fail to reflect the true quality of generated levels, as they are heavily biased towards high-frequency tokens. To address this, we propose a suite of metrics designed to explicitly quantify the fidelity and completeness of low-frequency tokens. Combined with existing evaluating methods, these metrics enable a comprehensive assessment of the generated levels. Finally, the fourth challenge lies in prompt engineering for LLMs. Even with context windows exceeding 1M tokens, generating large-scale 3D token-based level remains a demanding task that requires a well-crafted prompting strategy. A comprehensive record of our prompt design and the associated engineering workflow is available in the Appendix.

Experimental results demonstrate that our model is capable of generating levels with superior structural coherence and rich semantic details. The main contributions of this work are summarized as follows:

- We propose a Voxel VAE, a dense token embedding method that leverages multi-scale dilated convolutions to capture spatial context, yielding superior performance in downstream generative tasks compared to previous methods.
- We propose a Dual-Phase Curriculum Learning strategy, a diffusion training method designed to strike a balance between global structural consistency and detail reconstruction. This strategy partitions the training process into two phases: employing a standard MSE loss in the first phase, and progressively integrating ICF weighting into the loss function during the second phase.
- We introduce a set of metrics specifically tailored to evaluate the reconstruction fidelity of low-frequency tokens. In conjunction with existing metrics, this suite enables a comprehensive evaluation covering global structural consistency, diversity, and detail richness.
- We introduce CRAFT, a diffusion-based framework integrate a Voxel VAE with Dual-Phase Curriculum Learning. Extensive experiments demonstrate that CRAFT outperforms baselines, including modern LLMs utilizing sophisticated prompt engineering.

II. Related Work

This study relates to Procedural Content Generation (PCG), single-example generation,VAE and curriculum learning. In this section,we briefly survey the research in these domains.

A. PCG

PCG entails generating game content algorithmically to reduce manual workload [10]. Although traditional approaches—such as search-based and grammar-based methods—have proven effective, they are often bottlenecked by the requirement for extensive manual engineering. To overcome this issue, we situate our work within the framework of PCG via Machine Learning (PCGML) [11], [12], which leverages deep learning-based generative models to learn directly from existing contents.

Significant research efforts in both the commercial game industry and academic communities have been directed towards the automated generation of playable game contents. To address this challenge, prevalent methodological choices typically include GANs, VAEs, Reinforcement Learning (RL) and LLMs. Regarding GANs, researchers have employed this architecture to generate content encoded as pixel-based images or discrete tile arrays, as exemplified by applications in Super Mario [13] and Doom [14]. Hald et al. [15] employed a conditional GAN to generate levels for the puzzle game Lily's Garden. Gandikota and Brown [16] utilized a Deep Convolutional GAN to generate digital art. Another popular approach is using VAE. These models map levels into a continuous latent space, which is useful for blending different level styles [17]. To ensure the levels are actually playable, some researchers use RL. For example, the PCGRL framework [18] formulates level generation as a game where an agent learns to place tiles to maximize a quality score. Koya et al. [19] proposed a method to generate levels tailored to specified difficulty targets, employing a reward function based on completion time estimates. Recently, the application of LLMs into PCG has given rise to the text-to-level paradigm. A pioneering work, MarioGPT [20], treats level generation as a sequence modeling task, utilizing a fine-tuned GPT-2 to generate 2D levels from flattened token sequences. UnrealLLM [21] extends this methodology to 3D levels by directing LLMs to generate executable API scripts for game engines, thereby enabling the generation of complex levels. More significantly, the emergence of advanced foundation models, such as Gemini-3 [22] and GPT-5 [23], has demonstrated remarkable capabilities in few-shot learning and spatial reasoning. These advancements suggest a potential shift toward a training-free text-to-level paradigm, where 3D levels can be generated directly from prompts.

Our research is dedicated to token-based level generation [24], using Minecraft as a representative case for 3D games. We selected Minecraft not only for its large vocabulary of tokens and high semantic complexity but also for its vast repository of community-created levels and robust open-source ecosystem that facilitates direct access to in-memory game states and level data, enabling efficient model training and evaluation. Therefore, beyond level generation, Minecraft has recently emerged as a premier testbed for general artificial intelligence, fostering extensive research in open-ended embodied agents [25], [26], multi-agent collaboration [27], and complex planning with LLMs [28], [29]. Awiszus et al. [7] proposed World-GAN, an architecture similar to SinGAN [30] that learns from a single level in an unsupervised way. Sudhakaran et al [31] adopted Neural Cellular Automata (NCA) [32] in reconstructing Minecraft artefacts. Merino et al. [33] utilized interactive evolution alongside latent variable evolution for the procedural generation of target Minecraft structures. Diffusion models, such as DiffusionCraft [4], offer a better way to generate high-quality levels by removing noise from a gaussian noise input. Scaffold Diffusion [34] employs a masked discrete diffusion model conditioned on binary occupancy map to generate sparse structures.

B. Single-Example Generation

Data scarcity remains a critical limitation in deep learning-based PCG. While curated datasets like the Video Game Level Corpus (VGLC) exist for classic titles [35], practical game development often relies on a limited number of prototype levels. To address this, single-example generation aims to learn the internal patch distribution of a specific reference level rather than a general domain distribution [4]. While traditional statistical methods like Markov Chains [36] and WaveFunctionCollapse (WFC) [37] effectively capture local patterns, they fundamentally lack the capacity to model long-range dependencies required to generate coherent levels. To address this, GANs introduced a new paradigm. Inspired by SinGAN, Awiszus et al. proposed TOAD-GAN [6], employing a multi-scale pyramid architecture to generate token-based 2D levels of arbitrary sizes. This approach was later extended to 3D levels in World-GAN [7], which leveraged token embeddings to capture semantic relationships across a vast vocabulary of tokens. Despite their prevalence, single-example GANs often suffer from error accumulation due to multi-stage generation process, leading to structural artifacts [8]. Recently, Denoising Diffusion Probabilistic Models (DDPMs) have emerged as a robust alternative. Building on the single-scale architecture of SinDiffusion [8], DiffusionCraft [4] proposed the first diffusion-based framework for single-example level generation. This method introduces Block2Vec to map discrete tiles into continuous embeddings and utilizes a denoising network with a constrained receptive field. This design forces the model to learn local structural rules without overfitting.

C. Latent Diffusion Models (LDMs)

Since being proposed by Kingma and Welling [38], VAEs have transcended their initial role as merely generative models. From the perspective of representation learning, the inference network of a VAE is highly regarded for its ability to learn a latent space that is continuous, smooth. VAEs introduce Kullback-Leibler (KL) divergence regularization to constrain the distribution of latent variables to approximate a prior distribution. This mechanism endows the latent representation with probabilistic significance and robustness. This capability to learn robust and compact representations has facilitated the integration of VAEs with diffusion models, giving rise to LDMs. By performing the diffusion process in the compressed latent space, models achieve significantly improved computational efficiency and generation quality. This paradigm has been widely adopted in state-of-the-art text-to-image generation methods, including DALL·E [39], Stable Diffusion [40], and Flux [41]. This paradigm has also been successfully extended to the domain of 3D data generation. Lt3sd [42] introduced a latent tree representation method for diffusion process to generate large-scale 3D scenes. In the field of medical image synthesis, 3D MedDiffusion [43] employs a Patch-Volume Autoencoder to compress volumetric data into a latent space via patch-wise encoding, enabling efficient reconstruction through volume-wise decoding. Inspired by these pioneering works, we extend the LDM paradigm to the domain of single-example token-based level generation.

D. Curriculum Learning

Humans and animals learn much better when the examples are organized in a meaningful order which illustrates gradually more concepts, and gradually more complex ones. In the context of machine learning, such a training strategy is termed "curriculum learning" [44]. This paradigm has been widely adopted across various domains. For instance, Wang et al. [45] employed a curriculum weighting strategy in social recommender systems. Lange et al. [46] enhanced state-of-the-art method for Aspect-Based Sentiment Analysis (ABSA) by incrementally incorporating more difficult instances into the training process. In the realm of image generation, DisCL [47] integrated diffusion models with curriculum learning by adjusting the image guidance level at each training stage to improve the learning of hard examples. Li et al. [48] employed curriculum learning to effectively alleviate the training collapse caused by class imbalance in medical image segmentation.

III. METHOD

In this section, we present our proposed method. First, we introduce a Voxel VAE designed to map discrete 3D levels into a continuous latent embedding. Our Voxel VAE integrates a multi-scale dilated convolutional encoder without downsample as well as a context-aggregating decoder, incorporating ASPP [49] and SEBlock [50]. Second, we introduce a Dual-Phase Curriculum Learning strategy to train the DDPM within the latent space. The backbone of our DDPM follows the architecture established in DiffusionCraft [4]. The overview of our method is in Fig.1.

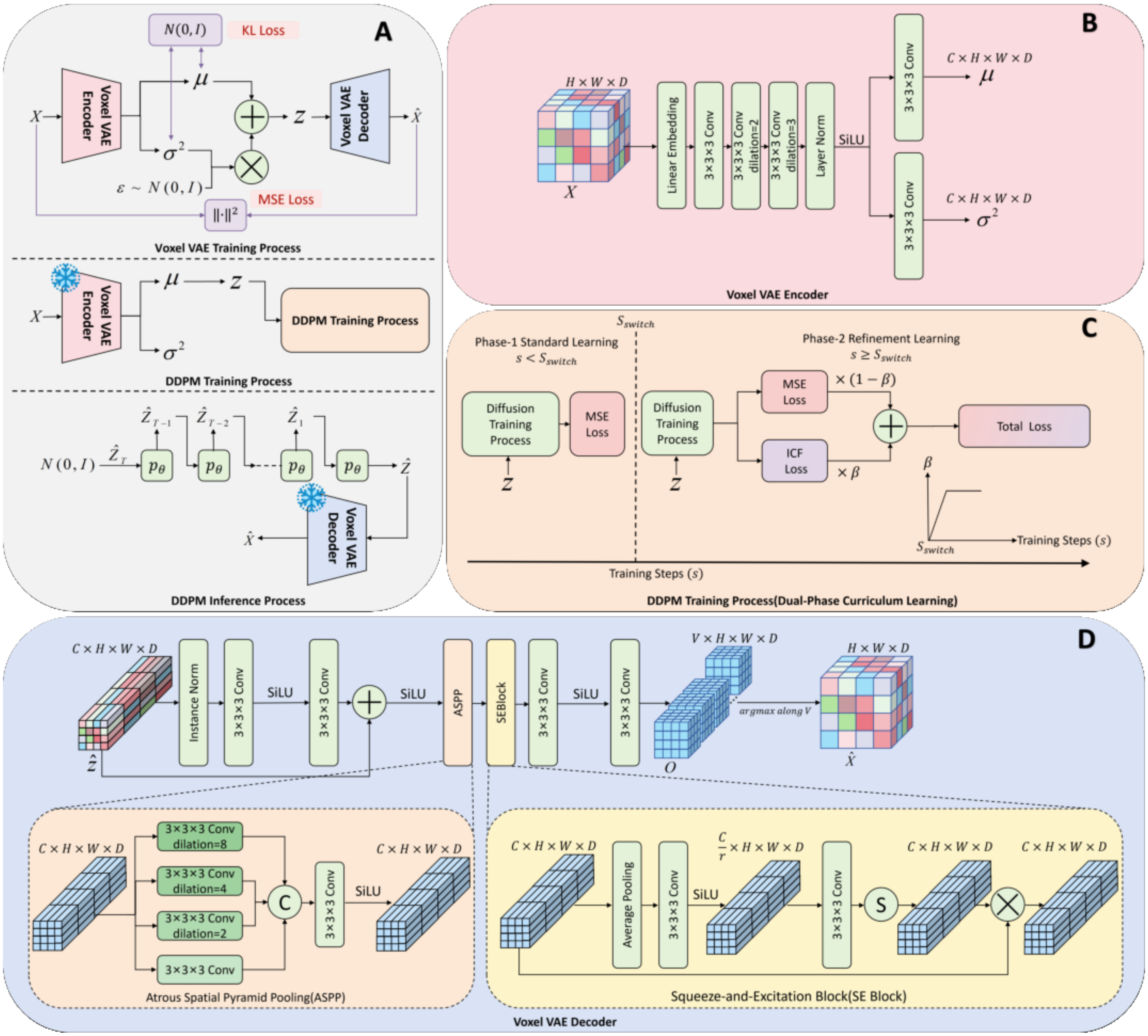

Fig. 1. Framework of CRAFT. The framework integrates Voxel VAE with Dual-Phase Curriculum Learning training strategy. (A) Overview of CRAFT. (B) Architecture of Voxel VAE Encoder. The Voxel VAE implements a multi-scale dilated convolutional encoder without downsampling to preserve spatial resolution. (C) The Dual-Phase Curriculum Learning strategy. This approach balances global structure learning and the reconstruction of low-frequency tokens by progressively integrating ICF weighting. (D) Architecture of Voxel VAE Decoder. The decoder incorporates ASPP and SEBlocks to enhance high-fidelity reconstruction by capturing multi-scale contextual information and adaptively reweight channel-wise features. Operations: ⓒ(Concatenation), ⊕(Element-wise Addition), ⊗ (Element-wise Multiplication), ⓢ(Sigmoid)

A. Voxel VAE

Previous works utilized dense fixed-size vectors to represent tokens by adopting the Block2Vec method, an approach analogous to word2vec in NLP.

Specifically, a 3D token-based level, denoted as X, comprises discrete tokens $t \in V$, resulting in a layout $X = [t_{ijk}]_{H\times W\times D}$. Here, H, W, and D denote the number of tokens along the respective dimensions, and V—analogous to the vocabulary in NLP—represents the set of all token types within the level. Each token t is associated with a semantic label consisting of one to three words (e.g., “Grass Block”). The embedding function $\mathrm{EMB}(t_{ijk})$ maps each token to a vector in $\mathbb{R}^C$. Consequently, the embedding of a level X of size $H \times W \times D$ is defined as $\mathrm{EMB}(X) = [\mathrm{EMB}(t_{ij})] \in \mathbb{R}^{CHWD}$.

However, Block2Vec outputs unsatisfactory embeddings under the long-tailed token distribution. Because the method relies solely on statistical co-occurrence, it struggles to capture meaningful semantic relationships among low-frequency tokens. Moreover, it ignores spatial context: identical tokens at different locations often carry different semantics. For example, water in a swimming pool and water in an ocean should be represented differently, but Block2Vec maps them

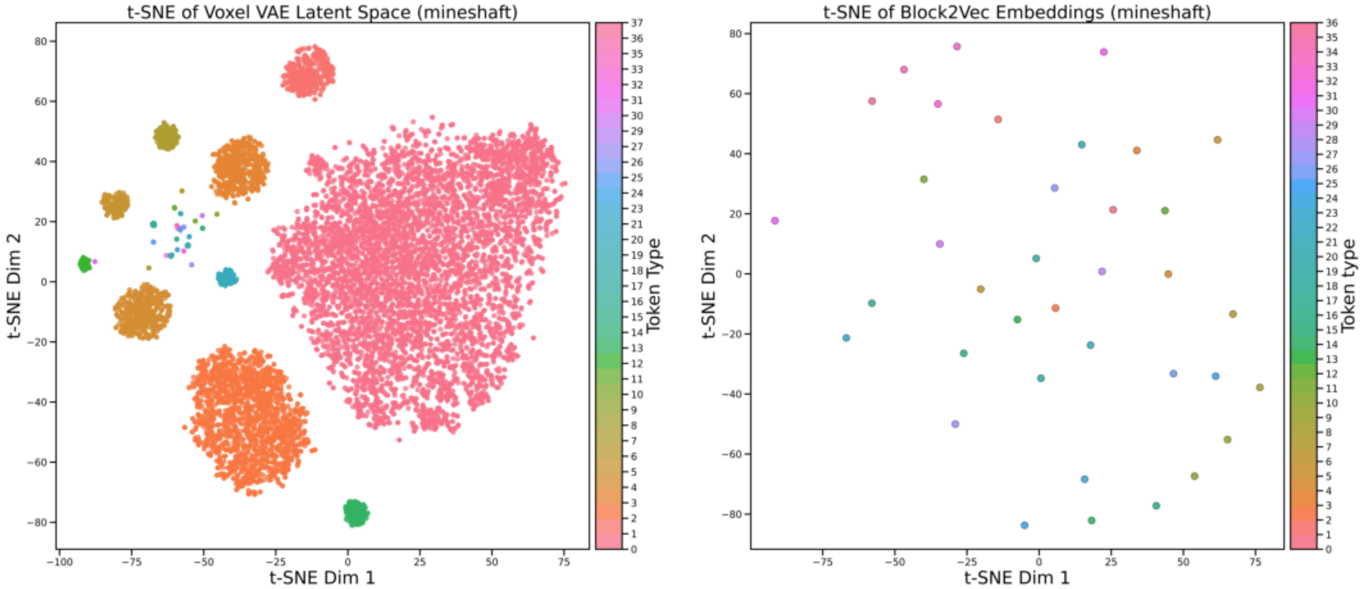

Fig. 2. Embeddings visualization for a specific level. We project the embeddings of tokens from each position in the original level into a 2D space, with points color-coded by their respective token types. As shown in the right panel, tokens of the same type cluster into singular, precise points, indicating that these embeddings are solely determined by the token type. Conversely, in the left panel, points of the same color are distributed across various locations. This is attributed to the Voxel VAE, which conditions embeddings on both token type and spatial context, effectively resolving semantic ambiguity.

into a same vector. To overcome these limitations, we replace Block2Vec with Voxel VAE that encodes with spatial context information and decodes by leveraging spatial context and channel-wise feature re-weighting to maximally reconstruct low-frequency tokens. The visualization of learned representations of Block2Vec and Voxel VAE is shown in Fig. 2.

1) Structure-preserving Encoder: Unlike traditional encoders that rely on downsampling operations to compress dimensions, our encoder operates without downsampling. This design is crucial not merely to minimize reconstruction error, but more critically, to preserve the spatial dimensions between the input level and the latent embeddings. By maintaining spatial dimensions, we establish a strict one-to-one correspondence where the latent variable at coordinate (i, j, k) is a direct projection of the variable at the same location in input level. This coordinate-wise alignment is imperative, as it allows us to directly associate the semantic class of each token with its latent representation, thereby enabling the computation of the ICF weighted loss during the diffusion training process. Consequently, instead of pooling, it employs a stack of dilated convolutions with varying dilation rates to expand the receptive field. This architecture allows the model to capture multi-scale context while preserving the original spatial resolution.$(H \times W \times D)$. In training phase, the original input X is mapped into a latent space and processed to predict the mean μ and variance σ^2 of the latent distribution. Subsequently, to enable differentiable sampling via the reparameterization trick [38], we derive the latent embedding z. Specifically, we sample a random noise tensor ϵ from a standard Gaussian distribution $\mathcal{N}(0, I)$. The final latent embedding is then computed as:

$$z = \mu + \sigma \odot \epsilon \tag{1}$$

where $\odot$ denotes the element-wise product. This stochastic sampling process is crucial during the training phase as it allows gradients to backpropagate through the random node [38]. To constrain the latent distribution to approximate the standard Gaussian prior, we incorporate the KL divergence into the loss function. The regularization term is formulated as:

$$\mathcal{L}_{KL} = -\frac{1}{2} \sum \left(1 + \log(\sigma^2) - \mu^2 - \sigma^2\right) \tag{2}$$

where the summation is performed over all latent dimensions. This ensures the learned manifold remains smooth and continuous [38].

In the inference phase, or when extracting latent embeddings to train the downstream diffusion model, we prioritize determinism to ensure a stable and consistent representation. Consequently, we bypass the stochastic sampling and directly utilize the predicted mean μ as the latent embedding:

$$z = \mu \tag{3}$$

This deterministic mapping provides the latent embeddings for the diffusion model training process.

2) Context-Aggregating Decoder: The decoder is tasked with reconstructing the discrete tokens from the continuous latent embedding. To accurately recover low-frequency tokens and ensure global structural consistency, we incorporate two key modules:

- **ASPP (Atrous Spatial Pyramid Pooling) [49]:** This module captures spatial structures at multiple scales, allowing the decoder to perceive the global structure of the level.
- **SEBlock (Squeeze-and-Excitation) [50]:** This module reweights channel-wise features, emphasizing informative features related to low-frequency tokens.

Algorithm 1 Dynamic Patch-based Training Strategy

```
Input: Single Level X, Voxel Budget V_budget, Interval I
while not converged do
    step ← step + 1
    if step (mod I) == 0 then
        r ← 1.0                              ▷ Periodic Full Sampling
    else
        Sample crop ratio r ~ U(0.1, 0.9)
    end if
    Calculate Patch Volume V_patch ≈ Vol(X) × r^3
    Batch Size B ← min(64, max(1, ⌊V_budget / V_patch⌋))
    X_batch ← ∅
    while |X_batch| < B do
        x_patch ← RandomCrop(X,r)
        if x_patch ∉ X_batch then
            X_batch ← X_batch ∪ {x_patch}
        end if                    ▷ Retry up to 5 times if duplicate
    end while
    Update Voxel VAE parameters φ, ψ using X_batch
end while
```

The decoder first maps the latent embedding z to a feature space through a series of convolutional layers. We then apply ASPP and SEBlock to enhance the feature representation and adaptively reweight channel-wise features. Finally, the decoder outputs a tensor O that represents a probability distribution over the token vocabulary, where dimension V denotes the vocabulary size. To reconstruct the discrete level $\hat{X}$, we take the argmax of the probability distribution O along dimension V.

3) Training Strategy: The Voxel VAE is trained end-to-end to maximize the evidence lower bound (ELBO) [38]. The total loss function combines the reconstruction loss and the KL divergence regularization:

$$\mathcal{L}_{VAE} = \mathcal{L}_{recon}(X, \hat{X}) + \lambda_{KL}\mathcal{L}_{KL} \tag{4}$$

where $\mathcal{L}_{recon}$ is the cross-entropy loss between the input level X and the reconstructed probabilities $\hat{X}$, and λ_{KL} is a weighting factor balancing reconstruction fidelity and latent space regularity.

To train the Voxel VAE on a single example, we adopt a patch-based strategy. Specifically, for each training batch, we sample a fixed crop size ratio $r \sim \mathcal{U}(0.1, 0.9)$ and extract random volumetric patches from the input level X with different spatial offsets. To prevent overfitting, we employ a rejection sampling strategy that discards duplicate patches, ensuring diversity within each batch. To maximize computational efficiency, we dynamically adjust the batch size inversely proportional to the patch volume. Additionally, we periodically sample the entire level as a batch of size 1 to ensure the model maintains global structural consistency. This approach ensures the Voxel VAE captures structures at various scales, as detailed in Algorithm 1. Here, V_{budget} represents the maximum processing capacity defined as a user-specified threshold strictly bounded by the available GPU memory, and I denotes the interval for periodic full-level sampling.

B. Dual-Phase Curriculum Learning Strategy

Standard DDPMs adopt MSE as the loss function; however, in our task this causes gradients dominated by high-frequency tokens, leading to the neglect of low-frequency tokens. To address this issue, we propose a two-phase learning strategy that first learns the global structure of the level and subsequently focuses on reconstructing low-frequency tokens, as detailed in Algorithm 2. Our generative backbone follows DiffusionCraft [4]: a 3D U-Net diffusion model with a limited receptive field. At the beginning of each training step, we apply a large random cropping operation with a crop size ratio 0.8 to the input level X to ensures that the model effectively captures the patch-level distribution [4]. To accelerate training and enhance visual fidelity, the loss function is formulated to directly predict the clean initial state [4] z_0 produced by the encoder of Voxel VAE. The training process is divided into two phases:

- **Phase-1 Standard Learning:** In this phase, we employ the MSE loss to enable the model to learn the global structure of levels.
- **Phase-2 Refinement Learning:** In this phase, we adopt a linear combination of MSE and ICF, where the ICF weight gradually increases from 0 to 1, as the loss function to focus the model on reconstructing low-frequency tokens.

Standard learning is consistent with the standard DDPM training process, with the loss function as follows:

$$\mathcal{L}_{std} = \|z_0 - \hat{z}_\theta(z_t, t)\|^2 \tag{5}$$

where z_0 is the clean initial state produced by the encoder of Voxel VAE, z_t is the noised state at time step t, and $\hat{z}_\theta(z_t, t)$ is the model's prediction. When the training steps exceed a hyperparameter S_{switch}, the phase switches to the refinement learning phase, with the loss function as follows:

$$\mathcal{L}_{ref} = (1 - \beta)\|z_0 - \hat{z}_\theta(z_t, t)\|^2 + \beta L_{icf} \tag{6}$$

where β is the ICF weight gradually increases from 0 to 1.

ICF re-weighting originates from the class-imbalance literature in computer vision [51], [52] and NLP [53], where it was first adopted to up-weight rare categories during supervised classification so that minority classes receive sufficient gradient signal. We adapt this idea to single-example generative modeling: instead of balancing a classifier, we balance the diffusion loss itself, giving low-frequency tokens a larger penalty when they are misreconstructed and thus forcing the generative network to preserve them. To do this, we introduce a re-weighting scheme based on ICF. We define the weight for a token type c as:

$$w_c = \min\left(\frac{N_{total}}{N_c}, \tau\right) \tag{7}$$

where N_{total} is the total number of tokens in the input level$X$, N_c is the frequency count of token type c, and τ is a clipping threshold to prevent gradient explosion. Since the shapes of both the original latent variable z_0 and the predicted latent variable $\hat{z}_\theta$ are $C \times H \times W \times D$, we weight the reconstruction error at every spatial location according to the token type

Algorithm 2 Dual-Phase Curriculum Learning Training

Input: Single Level X, Switch Step S_{switch}
$s \leftarrow 0$
repeat
  $s \leftarrow s + 1$
  $z_0 \leftarrow E_\phi(LargeRandomCrop(X))$
  Sample $t \sim \text{Uniform}(1, \dots, T)$
  Sample $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  $z_t \leftarrow \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$
  $\mathcal{L}_{std} \leftarrow \|z_0 - \hat{z}_\theta(z_t, t)\|^2$
  if $s < S_{switch}$ **then**
    $\mathcal{L} \leftarrow \mathcal{L}_{std}$ ▷ Phase-1
  else
    $\alpha \leftarrow \min(1, \frac{s - S_{switch}}{S_{switch}})$
    Calculate $\mathcal{L}_{icf}$ ▷ ICF Loss
    $\mathcal{L} \leftarrow (1 - \alpha)\mathcal{L}_{std} + \alpha\mathcal{L}_{icf}$ ▷ Phase-2
  end if
  Take gradient descent step on $\nabla_\theta \mathcal{L}$
until converged

occupying that location in the input level X. Concretely, we first obtain the token-to-index mapping $\mathcal{M} : (i, j, k) \mapsto t$ from the input level X, where $t \in V$ is the token type at location (i, j, k). Then, for each element in latent embedding indexed by (i, j, k), we assign the ICF weight w_c corresponding to token $c = \mathcal{M}(i, j, k)$. Subsequently, we construct the spatial weight matrix $\mathbf{W} \in \mathbb{R}^{H \times W \times D}$ by assigning w_c to each corresponding coordinate. The ICF-weighted loss is then computed as:

$$\mathcal{L}_{icf} = \|\sqrt{\mathbf{W}} \odot (\mathbf{z}_0 - \hat{\mathbf{z}}_\theta)\|^2 \tag{8}$$

where $\sqrt{\cdot}$ denotes the element-wise square root operation.

IV. Experiments

A. Datasets

For Minecraft level generation, the dataset comes from a large handcrafted Minecraft world called Drehmal: PrimΩrdial [54]. The levels included in the training data are shown in Table I. To assess the model's performance on level with extreme long-tail distribution, we manually constructed a level named 'Skyblock'. This level comprises 20,280 tokens across 15 types, with a single token type dominating approximately 96% of the levels. Furthermore, to better simulate real game scenarios, we distinguish tokens based on their attributes; game entities sharing the same type but differing in attributes such as orientation or switch state are treated as distinct token types. Consequently, this leads to a substantial increase in the number of token types within the training levels, exacerbating the long-tailed distribution and thereby demanding a significantly stronger capability from the model to reconstruct low-frequency tokens.

B. Evaluation Metrics

To quantitatively evaluate the quality and diversity for generated levels, we adopt the following metrics. Previous studies primarily relied on Tile Pattern KL-Divergence (TPKL-Div) to assess fidelity of generated levels. However, this metrics are heavily dominated by high-frequency patterns and often fail to capture the loss of patterns made up of low-frequency tokens. To address this limitation and provide a more comprehensive evaluation, we introduce novel metrics.

TABLE I
Statistics of the training levels used in our experiments.

Level	Size ($H \times W \times D$)	Vocabulary Size
Plains	$83 \times 23 \times 74$	54
Swamp	$49 \times 28 \times 52$	72
Mineshaft	$42 \times 18 \times 45$	38
Skyblock	$13 \times 130 \times 12$	15
Village	$40 \times 20 \times 34$	100

a) Tile Pattern KL-Divergence (TPKL-Div): To measure the local patterns similarity between the generated levels and the original level, we employ Tile Pattern KL-Divergence [55]. The formula is as follows:

$$D_{KL}(P||Q) = \sum_{s \in S_P} P'(s) \log\left(\frac{P'(s)}{Q'(s) + \epsilon}\right) \tag{9}$$

where S_P denotes the collection of all unique patterns. $P'(s)$ refers to the frequency of pattern s in the training level and $Q'(s)$ refers to its frequency in the generated level. ϵ is a small constant set to 10^{-7} to prevent division by zero when a pattern s exists in the training level but is missing in the generated levels. We use three tile pattern windows (3×3, 5×5, 10×10) to calculate the TPKL-Div. We measure two kinds of TPKL-Div between the training level and generated level distributions and sum them up with a weight $w = 0.5$. As indicated by Equation 9, the contribution of each pattern s to the total divergence is weighted by its frequency $P'(s)$. Consequently, low-frequency patterns—typically those containing low-frequency tokens—have a negligible impact on the final metric compared to high-frequency patterns. This implies that TPKL-Div is dominated by high-frequency patterns and primarily reflects the global structural consistency.

b) Levenshtein Distances(LD): In order to assess the diversity of generated samples, we adopt the Levenshtein [56] distance as a measure to quantify the differences between training level and generated levels. The Levenshtein distance is a standard metric for quantifying the difference between two sequences. It is defined as the minimum number of single-character edits—specifically insertions, deletions, or substitutions—required to transform one sequence into the other. In this work, we flatten both the training and generated levels into one-dimensional sequences and encode each token as a numerical index to calculate Levenshtein distance.

c) Global Token Frequency(GTF): Previous works have also focused on evaluating the generation of low-frequency tokens. For example, World-GAN [7] proposed using frequency histograms to assess the generation of these low-frequency tokens. However, such methods lack quantitative precision for direct comparison.

To address this issue, we adopt the Tile Pattern KL-Divergence with a tile pattern window size of 1×1. Mathematically, a 1×1 pattern corresponds to a single token.

Therefore, computing 1×1 TPKL-Div effectively measures the divergence between the global token frequency distributions of the generated level and the training level. Moreover, it serves as a crucial indicator of whether the ICF weighting is over-corrected, which could cause the model to compromise the overall token frequency distribution in its effort to learn low-frequency tokens.

d) Rare Density Error (RDE): Although the Global Token Frequency metric effectively assesses whether the model has learned the global token distribution of the training level, it remains susceptible to being dominated by high-frequency tokens. To rigorously quantify the deviation in low-frequency tokens, we employ the Rare Density Error. Since low-frequency tokens occupy a negligible proportion of the level , measuring absolute difference is insufficient. Instead, we calculate the relative error to assess how well the model reconstruct low-frequency. We define low-frequency as tokens with a frequency below a threshold $\tau_{RDE} = 1\%$ in the training sample. Let $\rho_{generated}$ and ρ_{train} denote the density of these low-frequency tokens in the generated and training levels. The RDE is formulated as:

$$E_{rare} = \frac{|\rho_{generated} - \rho_{train}|}{\rho_{train}} \tag{10}$$

A lower RDE indicates that the generative model exhibits a superior ability to reconstruct low-frequency tokens.

e) Category Coverage Rate(CCR): While Rare Density Error measures the ability of model to reconstruct low-frequency tokens, it does not reflect the completeness of the generated vocabulary. To address this issue, we propose the Category Coverage Rate.Let $\mathcal{V}_{train}$ denote the set of token types present in the training level, and $\mathcal{V}_{generated}$ denote the set of token types appearing in a generated level. The CCR is computed as the intersection ratio:

$$\text{CCR} = \frac{|\mathcal{V}_{generated} \cap \mathcal{V}_{train}|}{|\mathcal{V}_{train}|} \times 100\% \tag{11}$$

where $|\cdot|$ denotes the cardinality of the set. A higher CCR indicates that the generative model can reconstruct more token types of the training level.

C. Implementation Details

The proposed framework is implemented using PyTorch, and all experiments were conducted on a single NVIDIA RTX 3090 GPU. The Voxel VAE utilizes the SiLU activation function. For the latent diffusion model, we set the maximum timestep to $T = 1000$. The model is trained for $100,000$ iterations with a batch size of 8, using an Exponential Moving Average (EMA) decay of 0.9999 to stabilize training. In the Dual-Phase Curriculum Learning strategy, the phase transition occurs at step $S_{\text{switch}} = 40,000$, with a clipping threshold $\tau = 5$. The initial learning rate is set to 5×10^{-4}. During the Refinement Learning phase, we employ a cosine annealing schedule to decay the learning rate to 5×10^{-5}. To prevent gradient explosion, we employ gradient clipping with a maximum norm of 1.0.

D. Comparison with Previous Methods

To comprehensively evaluate our method, we compare it against three categories of approaches: GAN-based methods (World-GAN), Diffusion-based methods (DiffusionCraft) and Foundation Model-based approaches. For the latter, we select Gemini-3 as the representative baseline. As one of the most advanced Multimodal Large Language Models (MLLMs), Gemini-3 demonstrates exceptional capabilities in spatial reasoning [57], allowing it to generate 3D voxel structures via prompting without task-specific training. Specifically, we serialize the single reference level into a textual sequence. This sequence is provided as part of the prompt, instructing the model to capture the structural patterns and generate a novel level. The detailed prompt engineering and the full text of the prompt used for Gemini-3 can be found in Appendix. For World-GAN and DiffusionCraft, which are established methods in the PCG domain, we use the hyperparameter settings recommended in their original publications to ensure a fair comparison. For each method, we generate 20 random samples and calculate the average evaluation metrics.

The quantitative results are shown in Table II. Although our method may underperform baseline methods on a single metric in certain scenes, it demonstrates a clear overall advantage when all scenes and metrics are considered. This is primarily attributed to our Dual-phase Curriculum Learning, which balances global structural consistency with low-frequency token reconstruction, yielding superior robustness across diverse scenarios.

To complement these objective metrics with perceptual evaluation, we visually compare the generated levels in Fig. 3. In the village scene, the zoomed-in region represents a critical structure made up of low-frequency tokens. The level generated by World-GAN tends to exhibit artifacts. Although DiffusionCraft produces coherent structures, it fails to reconstruct the critical structure made of low-frequency tokens and has less detail compared to the training level. Gemini-3 fails to preserve a clear global structure. In contrast, our method generates levels that are not only structurally coherent but also rich in detail. In the Skyblock scene, characterized by a tree centered on an island surrounded by air, previous methods struggle due to the extreme long-tailed distribution of tokens. Consequently, both DiffusionCraft and Gemini-3 fail to generate a clear global structure. While World-GAN manages to produce a coherent structure, it generates an unrecognizable semantic structure. In contrast, although our generated levels exhibit slightly reduced variety compared to the training data, our method successfully yields high-fidelity results.

To effectively evaluate the playability of the generated levels, we conducted a user study involving 35 experienced players. The survey presented visualizations of levels generated by different models for the Village and Skyblock scenes. Participants ranked the four models based on three questions: Q1: Which result has the best structural completeness and architectural logic? Q2: Are the details rich and do they appear natural? Q3: Which level is the most appealing to play in? The mean ranking results are summarized in Table III. According to the results of the user study, our model produces level that

TABLE II
QUANTITATIVE EVALUATION. ↑: A HIGHER METRIC VALUE IS BETTER; ↓: A LOWER METRIC VALUE IS BETTER. **BOLD** SCORES ARE THE BEST RESULTS. THE **GTF** VALUES ARE SCALED USING SQUARE ROOT TRANSFORMATION ($\sqrt{\cdot}$) FOR BETTER READABILITY.

Scene	Method	TPKL-Div ↓	LD ↑	CCR (%) ↑	RDE (%) ↓	GTF ($\sqrt{\cdot}$) ↓
plains	World-GAN (ToG)	25.74	46220	22.96	57.83	0.303
	DiffusionCraft (AAAI)	**22.74**	40066	37.31	31.88	0.291
	Gemini-3	33.33	**66605**	**87.03**	**3.48**	0.322
	Ours	23.18	43642	84.99	17.61	**0.130**
swamp	World-GAN (ToG)	27.34	12474	39.65	**3.06**	0.458
	DiffusionCraft (AAAI)	**19.75**	13658	62.36	43.21	0.533
	Gemini-3	27.16	23183	70.83	67.31	0.401
	Ours	21.48	**14137**	**84.44**	27.42	**0.381**
mineshaft	World-GAN (ToG)	41.11	8228	59.34	57.83	0.306
	DiffusionCraft (AAAI)	32.85	11095	83.68	50.13	0.232
	Gemini-3	37.60	10936	81.57	**25.06**	**0.226**
	Ours	**29.88**	**13496**	**97.36**	36.61	0.251
village	World-GAN (ToG)	35.41	7347	35.25	85.13	0.303
	DiffusionCraft (AAAI)	28.09	10298	72.24	30.19	0.238
	Gemini-3	37.80	**15263**	58.00	**7.92**	0.481
	Ours	**27.95**	14436	**82.95**	16.71	**0.215**
skyblock	World-GAN (ToG)	3.11	370	92.00	26.27	**0.101**
	DiffusionCraft (AAAI)	**1.51**	726	38.33	79	0.430
	Gemini-3	31.77	**10206**	**100.00**	1525.86	0.891
	Ours	1.67	420	63.66	**13.73**	0.171

TABLE III
MEAN RANK OF USER STUDY RESULTS. PARTICIPANTS RANKED THE METHODS FROM 1 (BEST) TO 4 (WORST). LOWER VALUES INDICATE BETTER PERFORMANCE. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

	Village			Skyblock		
Method	**Q1**	**Q2**	**Q3**	**Q1**	**Q2**	**Q3**
World-GAN	2.46	3.20	2.83	3.00	3.00	3.29
DiffusionCraft	2.77	2.60	2.94	2.43	2.89	2.91
Gemini-3	3.54	2.97	2.97	3.54	3.09	2.66
Ours	**1.23**	**1.23**	**1.26**	**1.03**	**1.03**	**1.14**

is most favored by participants.

E. Ablation Study

This section presents comprehensive ablation studies to evaluate our key design choices, verifying the essential role of the Dual-Phase Curriculum Learning strategy and identifying the optimal hyperparameter.

Firstly, we conduct an ablation study to validate the effectiveness of design choices in our method. Specifically, we investigate the impact of the Voxel VAE and the Dual-phase Curriculum Learning strategy on the quality of generated levels. To isolate the contribution of each component, we compare our full model against variants where the Voxel VAE or the Dual-phase Curriculum Learning strategy is removed. We also evaluate a baseline model with neither component. The quantitative comparison, based on 20 generated levels, is presented in Table IV. The results demonstrate that the ICF weighting in the Dual-Phase Curriculum Learning strategy enables the model to effectively reconstruct low-frequency tokens, incurring only a negligible cost to overall structural similarity and diversity. Additionally, the Voxel VAE significantly improves the quality and diversity of generated levels, with only a negligible trade-off in structural similarity.

To validate the necessity of the Dual-Phase Curriculum Learning strategy, we conducted a comparative experiment against a single-phase baseline. We established three experimental groups, each trained for a total of 100, 000 steps and used Block2Vec as encoder. The first group serves as the baseline, utilizing the standard DiffusionCraft framework. The second group employs a single-phase training approach, where the ICF weighted loss is applied from the beginning. The third group employs our Dual-Phase Curriculum Learning strategy, utilizing standard MSE loss for the initial 40,000 steps and transitioning to the combined MSE and ICF loss for the subsequent 60,000 steps. The comparison is presented in Table V. The results demonstrate that while applying ICF loss from the beginning allows the diffusion model to reconstruct low-frequency tokens, it inevitably harms global structural consistency, as evidenced by the increased TPKL-Div. This suggests that early reweighting disrupts the learning of global structure. In contrast, our Dual-Phase Curriculum Learning strategy (Group 3) strikes a balance: it improves global structural consistency while effectively reconstructing low-frequency tokens. Specifically, our method not only achieves the most accurate reconstruction of low-frequency tokens—yielding the lowest RDE (39.59) and GTF (26.32)—but also

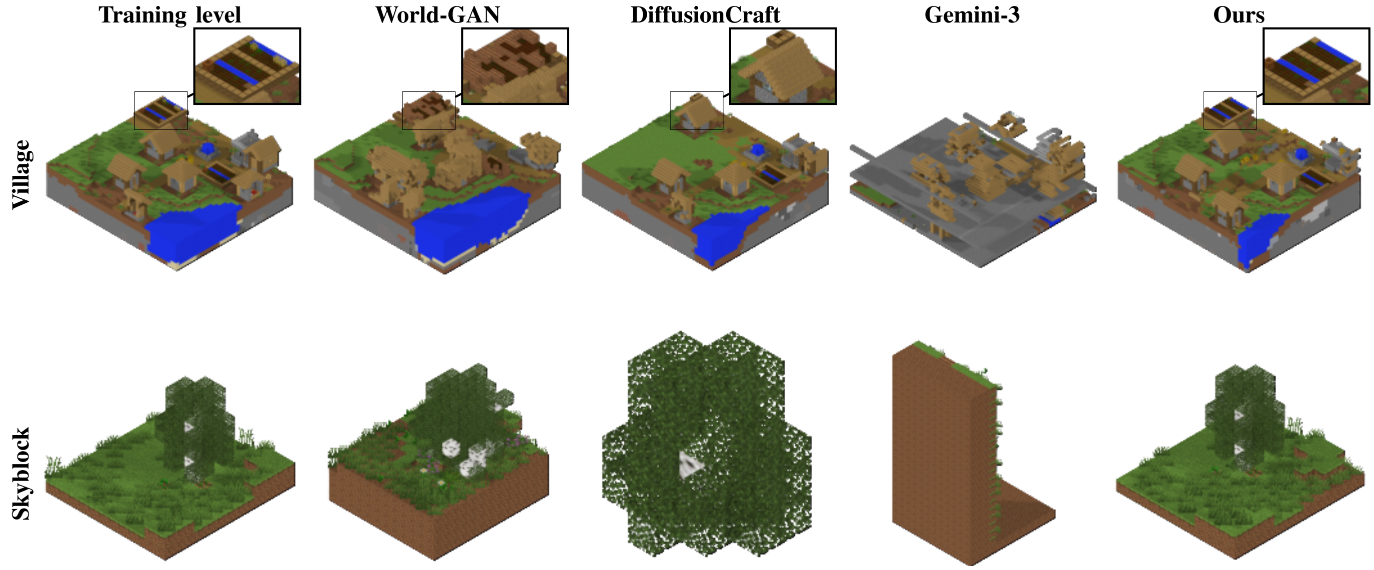

Fig. 3. Visual comparison across different scenes. Top Row: Results in the Village scene. Bottom Row: Results in the Skyblock scene. Notably, our model demonstrates superior capability in generating complex semantic structures, as evidenced by the undulating terrain, dense ground decorations, and the farmland (highlighted in the zoomed-in boxes) shown in the top row. Furthermore, our model exhibits remarkable robustness when trained on extremely sparse levels, as demonstrated by the coherent structural preservation in the bottom row.

TABLE IV
ABLATION STUDY ON KEY DESIGN CHOICES.

Method	VAE	ICF	TPKL-Div↓	LD↑	CCR(%)↑	RDE(%)↓	GTF($\sqrt{\cdot}$)↓
Baseline	×	×	28.09	10298	72.24	50.13	0.238
w/o ICF	✓	×	28.24	14315	77.64	18.24	0.218
w/o VAE	×	✓	28.32	10099	73.90	25.77	0.224
CRAFT (Ours)	✓	✓	**27.95**	**14436**	**82.95**	**16.71**	**0.212**

TABLE V
ABLATION STUDY ON DUAL-PHASE CURRICULUM LEARNING STRATEGY. ↑: A HIGHER METRIC VALUE IS BETTER; ↓: A LOWER METRIC VALUE IS BETTER. **BOLD** SCORES ARE THE BEST RESULTS

Group	Method	Training Schedule	Metrics				
			TPKL-Div↓	LD↑	CCR(%)↑	RDE(%)↓	GTF($\sqrt{\cdot}$)↓
1	Baseline	Standard (100k steps w/ MSE)	32.85	11095	83.68	50.13	0.232
2	Single-Phase	w/o Curriculum (100k steps w/ ICF)	33.11	11747	83.42	48.05	**0.223**
3	Dual-Phase (Ours)	0-40k (MSE) → 40k-100k (MSE+ICF)	**29.88**	**13496**	**97.36**	**36.61**	0.251

preserves the highest structural coherence with the lowest TPKL-Div (30.63). This confirms that decoupling Standard Learning from Refinement Learning is essential for high-fidelity level generation.

To investigate how hyperparameters regulate the trade-off between global structural consistency and low-frequency tokens reconstruction, we conducted a sensitivity analysis on the clipping threshold τ and the phase transition step S_{switch}. Table VI presents the impact of the clipping threshold τ. We observe that performance initially improves as τ increases, peaking around $\tau = 5$ and $\tau = 10$. This range effectively balances the learning of global structure and low-frequency details. Beyond this optimal range (e.g., $\tau = 20$), the model suffers from loss of structural coherence, evidenced by a sharp decline in TPKL-Div. Consequently, we select $\tau = 5$ as default, as it yields the most robust performance across all metrics.

TABLE VI
SENSITIVITY ANALYSIS OF CLIPPING THRESHOLD τ. WE EVALUATE THE IMPACT OF DIFFERENT CLIPPING THRESHOLDS IN THE DUAL-PHASE CURRICULUM LEARNING STRATEGY. ↑: A HIGHER METRIC VALUE IS BETTER; ↓: A LOWER METRIC VALUE IS BETTER. **BOLD** SCORES ARE THE BEST RESULTS

τ	TPKL-Div↓	LD↑	CCR(%)↑	RDE(%)↓	GTF($\sqrt{\cdot}$)↓
1	28.01	13041	78.50	26.89	0.247
3	28.26	13084	76.35	21.11	0.225
5	**27.95**	14436	**82.95**	16.71	0.215
10	28.28	13037	81.10	**11.13**	**0.213**
20	29.74	**16671**	72.90	29.23	0.377

TABLE VII
SENSITIVITY ANALYSIS OF PHASE TRANSITION STEP S_{SWITCH}. WE INVESTIGATE THE EFFECT OF THE TRANSITION STEP S_{SWITCH} IN THE DUAL-PHASE CURRICULUM LEARNING. ↑: A HIGHER METRIC VALUE IS BETTER; ↓: A LOWER METRIC VALUE IS BETTER. **BOLD** SCORES ARE THE BEST RESULTS

S_{switch}	TPKL-Div↓	LD↑	CCR(%)↑	RDE(%)↓	GTF($\sqrt{\cdot}$)↓
20000	28.53	13429	82.25	16.87	**0.190**
40000	27.95	**14436**	**82.95**	**16.71**	0.215
60000	28.10	13338	80.10	27.17	0.245
80000	**27.38**	13633	79.45	18.98	0.257

Following the analysis of τ, we investigate the impact of the phase transition step S_{switch}. Table VII details the model's performance under different transition schedules. We observe that an early switch ($S_{\text{switch}} = 20,000$) results in suboptimal structural consistency (higher TPKL-Div), likely due to insufficient training on global patterns. Conversely, extending the first phase too long ($S_{\text{switch}} \geq 60,000$) restricts the refinement of low-frequency tokens. Notably, while $S_{\text{switch}} = 80,000$ yields the best structural consistency (lowest TPKL-Div of 27.38), it hampers the capability to generate diverse and low-frequency patterns, evidenced by a worsened RDE and GTF. Our results indicate that $S_{\text{switch}} = 40,000$ achieves the best trade-off, as it ranks within the top two across all metrics, providing the best comprehensive performance.

F. Conclusion and Future Work

In this paper, we presented CRAFT, a novel framework designed for high-quality single-example level generation. To address the challenges of semantic sparsity and long-tailed distribution of tokens, we introduced a Voxel VAE that effectively resolves the semantic ambiguity of discrete tokens by capturing spatial context. Furthermore, our proposed Dual-Phase Curriculum Learning strategy successfully decouples global structure learning from the reconstruction of low-frequency tokens, enabling the model to reconstruct rare details without compromising global structural coherence. Extensive experiments on Minecraft scenes demonstrate that CRAFT significantly outperforms existing baselines, including GAN-based, diffusion-based, and foundation model-based approaches. The user study further validates the playability of our approach.

Despite these advancements, our approach maintains certain limitations. Strictly adhering to a single reference example may limit the emergence of novel structures not implicitly present in the training level. In future work, we will continue to explore single-example generation while investigating methods to incorporate MLLMs . By injecting prior knowledge into the generation process, we aim to facilitate the emergence of novel structures that are not found in the training example.

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